

# CIBIE CHARAN S

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## Objective

Seeking a position in Research & Development within IoT and Embedded Development, where I can apply my strong technical expertise and experience in knowledge-sharing to contribute to cutting-edge innovations and drive organizational progress

## Education

### Government College of Engineering, Srirangam

2023

BE Electronics and Communication Engineering

Trichy, Tamilnadu

### Mount Park Higher Secondary School

2018

Higher Secondary

Thiyagadurgam, Tamilnadu

## Experience

### Technical Coordinator | Bct Labs

Nov 2024 – Present

- Lead and coordinate a technical team for R&D projects, robotics development, and prototyping activities.
- Supervise the design and development of robotic systems integrating stepper motors, drivers, sensors, and control systems.
- Oversee rapid prototyping processes, including hardware testing, system integration, and product development workflows.
- Work with laser-based systems and motion control components in experimental and development projects.
- Utilize Bambu Lab P1S and Bambu Lab A1 3D printers for prototyping mechanical parts and functional components.

### Associate Robotics Engineer | Ashtrix Robotics and Research Center

Jan 2024 – Nov 2024

- Conducted R&D-focused training on Arduino, Raspberry Pi, and ESP8266, emphasizing practical hardware–software integration.
- Developed training modules on IoT communication protocols for advanced prototyping and real-world applications.
- Provided guidance on troubleshooting communication protocols to build strong problem-solving skills in IoT systems.
- Supervised hands-on IoT projects, helping students integrate hardware, software, and communication while promoting collaborative R&D-style learning.

### Production Trainee | KritiLabs Technologies, Chennai

July 2023 - Jan 2024

- Performed comprehensive PCB testing to ensure functionality and adherence to quality standards, contributing to the production of high-reliability electronic systems.
- PCB assembly and service processes, optimizing efficiency and ensuring the seamless integration of components in line with production goals.
- Diagnosed and troubleshooted PCB issues, applying root cause analysis techniques to resolve faults and improve overall product performance.

### Graduate Innovation Engineer | Forge Innovation and Ventures

Feb 2022 - Jun 2023

- Innovated as a Graduate Innovation Engineer at Forge Innovation and Ventures, specializing in rapid prototyping, Eagle PCB design, soldering, Embedded Systems, IoT, AutoCAD, UI/UX, and LabVIEW myRIO.
- Applied design thinking principles to develop user-centered solutions, showcasing a strong commitment to user experience and problem-solving.

## Projects

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### Smart Farming Redefined: Meet ROVER 4WDXL60R, the Weed-Eliminating Robot | *Bct Labs*

- Robust 4-Wheel Drive Rover – Built with a 1 mm metal chassis, 4WD system, and 4-wheel turning mechanism for smooth movement on different terrains with 5 kg payload capacity.
- AI-Based Weed Detection – Camera trained using PictoBlox AI detects weeds and activates a 24V focusable laser to eliminate them.
- Precision Motion System – Rail-guided stepper motor with TB6600 driver moves the laser and camera horizontally for accurate targeting.
- High-Power Motor Control – Uses BTS7960 43A motor driver for DC motors and 40KG S8218 servo for steering control.
- Smart Connectivity & Monitoring – Supports autonomous/manual control, FlySky remote operation, Bluetooth connection, siren alert on weed detection, and WiFi camera for live monitoring.

### Restoring Polluted Waters with HydroDrift's Innovative IoT Solution | *Bct Labs*

- Real-Time Water Monitoring – Uses pH, TDS, and turbidity sensors with a 4x4 I2C LCD display to continuously analyze and display water quality.
- Arduino R4 WiFi IoT Control – The Arduino R4 WiFi manages system operations and enables remote monitoring through IoT connectivity.
- Manual Navigation System – Controlled using a FlySky transmitter/receiver, BCT motor driver, and Nanobot controller for forward, backward, and turning movement on water.
- Efficient Power & Filtration System – Includes sediment filtration, relay-controlled motor pump, and regulated power supply using 7809 and 7805 voltage regulators for stable operation.

### Autonomous Lane Tracking | *Ashtrix Robotics and Research Center*

- Led the development of an "Omni Directional Wind Profiler" project, utilizing simulation and analysis tools for comprehensive system evaluation.
- Successfully integrated the Cayenne Cloud platform for efficient data visualization and real-time monitoring of the wind profiler.
- Demonstrated hands-on skills by building a functional prototype, showcasing practical application and implementation of the project.
- Optimized computational efficiency by fine-tuning OpenCV algorithms and leveraging Raspberry Pi 4's capabilities, ensuring reliable and responsive performance for real-time lane tracking and obstacle avoidance

## Technical Skills

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**Languages:** Python, C

**Microcontrollers:** Arduino, NodeMCU, ESP32, Raspberry Pi

**Softwares :** Arduino IDE, KiCad, Proteus

**IoT & PCB Design:** IoT Development, PCB Design and Troubleshooting

**3D Printing :** Bambu labs p1s, Bambu labs A1, Creality CR10 MAX, Adventure 4 series, Ender v3

## Languages

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- English
- Tamil